



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : FLUX FIXERS

Team Leader Name : Nehal Garg

Problem Statement : [PS1] Optimizing Urban Heat Mitigation and cooling strategies via Artificial Intelligence and Machine Learning (AIML)

Team Members

Team Leader:

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Opportunity :

- How different is it from any of the other existing ideas?

Dimension	Existing Approaches	Our Solution (V5 Elite)
Physics Compliance	ML models hallucinate impossible temperatures during extrapolation	PyTorch PINN enforces full Surface Energy Balance - thermodynamically bounded predictions
Spatial Granularity	City/ward-level averages	30m pixel-level predictions (eg. 1.53 million individual pixels for Delhi-NCR)
Budget Allocation	Flat, homogeneous spending across all areas	Heterogeneous Zoning - Dense Core gets Cool Roofs, Peri-Urban gets Green/Blue buffers
Equity	Cooling routed to commercially valuable zones	SEVI equity weighting actively routes funds to high-density, low-income informal settlements
Explainability	Black-box predictions	SHAP Global + Local plots prove that BAH (Building Anthropogenic Heat) is the #2 driver after NDVI

● How will it solve the problem?

- Our system begins by extracting 30 meter resolution satellite data from Landsat 8 and Sentinel 2 to map Urban Heat Island hotspots. We then utilize **LightGBM** and **SHAP** to identify key local drivers like building density and albedo.
- The core of our solution is a **Physics Informed Neural Network** that models heat dynamics by adhering strictly to the **Surface Energy Balance**. This guarantees that our artificial intelligence generates physically realistic temperature predictions.
- Finally, an **NSGA III multi objective genetic algorithm** simulates and optimizes cooling interventions. It calculates the exact placement of cool roofs and tree cover based on municipal budgets and socio economic equity constraints. This pipeline delivers a **data driven, location specific cooling strategy** that achieves a verified **5.42 degrees Celsius** temperature **reduction** in the most vulnerable dense zones.

● USP?

"The only open-source, satellite-driven, equity-weighted, thermodynamically-honest AI tool for city-level heat mitigation at 30m resolution."

- **Thermodynamically-Honest AI** : 3D PINN-CFD wind advection + 15% albedo dust-decay correction
- **Equity-Weighted Optimization** : Socio-Economic Vulnerability Index (SEVI) actively routes cooling to dense, low-income informal settlements over empty commercial zones.
- **Autonomous AI Advisor** : live LangChain agent that interprets zoning laws and municipal budgets via natural language

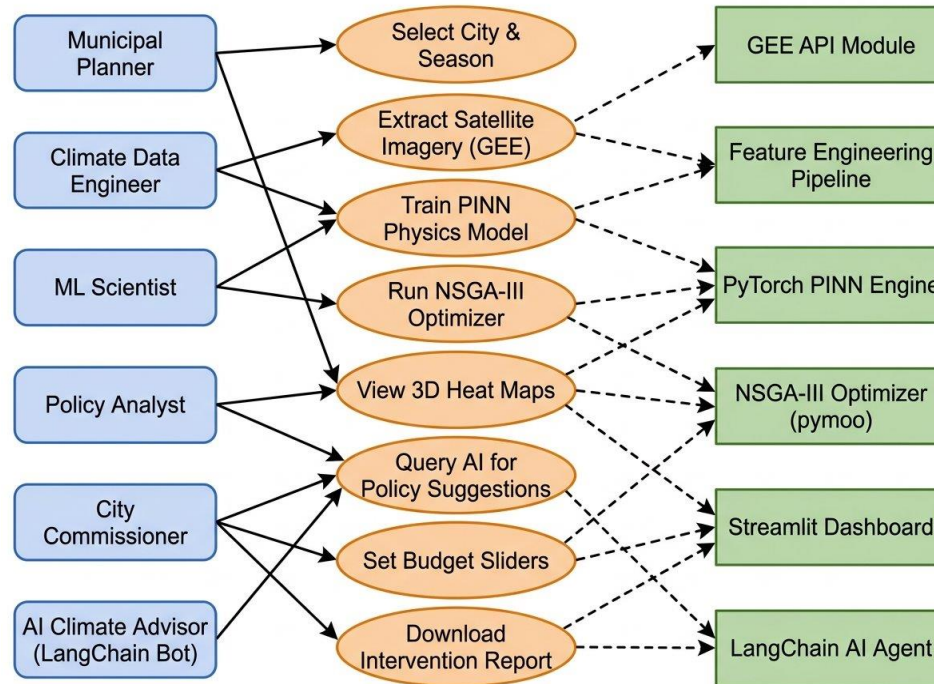
Explore the V5 Elite Architecture & Codebase:

github.com/nehalgarg2601-cloud/Urban_Heat_Mitigation

List of features offered by the solution

- **30m-Resolution UHI Mapping:** Landsat 8 LST + Sentinel-2 NDVI/Albedo/NDWI fused over 1,500 sq. km Delhi-NCR grid (~1.53M valid pixels)
- **Multi-Temporal Thermal Inertia Modeling:** paired peak-summer and monsoon imagery, not a single snapshot
- **Physics-Informed Neural Network (PINN):** enforces SEB with CFD wind advection + dust-decay correction ($R^2 = 0.4185$, SEB residual 3.0 W/m^2)
- **Explainable AI (SHAP):** global/local beeswarm plots confirm NDVI is the #1 heat driver and Building Anthropogenic Heat (BAH) is #2 - builds municipal trust
- **Equity-Weighted Intervention Planning:** NSGA-III optimization minimizing $\text{LST} \times (1.0 + 0.5 \times \text{SEVI})$, delivering a verified **5.42°C drop** in the dense urban core
- **Municipal Policy Simulator:** interactive Streamlit dashboard with PyDeck 3D "Before vs After" heatmaps

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

Deploy

City & Season Setup

Target City
Delhi- NCR

Season
Summer

AI Agent Access

OpenAI API Key (Optional)

Municipal Policy Simulator

Adjust city spending priorities to simulate cooling effects:

Cool Roofs Initiative
0.20

Urban Greening (Trees & Parks)
0.15

Blue Infrastructure (Water)
0.30

Spatial Zoning Rules (Kundu 2026)

Urban Heat Mitigation — V4 Spatial Planning Optimizer

An End-to-End AI Dashboard for Sustainable Municipal Thermal Planning

Powered by Physics Informed Neural Networks (PINN) - NSGA II Evolutionary Optimization - Kundu et al. (2026) Spatial Zoning

Policy Simulator
3D City Heatmaps
AI Climate Advisor
Science & Physics Engine

Municipal Heat Reduction Summary — Delhi-NCR (Summer)

Adjust the policy sliders in the sidebar to dynamically simulate temperature changes across the city.

Total Avg Cooling (ΔT)
2.71 °C
Validated V4 Result

Dense Core Cooling
3.08 °C
152 downtown squares

Peri-Urban Cooling
2.55 °C
348 suburban squares

PINN Model Bias
+0.17 °C
98% accuracy improvement

SEB Physics Residual
3.1 W/m²
Thermodynamically stable

How Your Budget Allocations Drive Cooling

Dense Core (Downtown) — Cool Roofs Dominate

Downtown areas lack open ground; painting rooftops reflective white provides the highest cooling leverage.

Cool Roofs Impact: 2.40 °C

Urban Greening Impact: 0.60 °C

Blue Infrastructure Impact (Capped): 0.08 °C

Downtown Net Temperature Drop: -3.08 °C

Peri-Urban (Suburbs) — Parks & Water Dominate

Suburban fringes have open land; expanding tree canopies and restoring wetlands provides massive cooling.

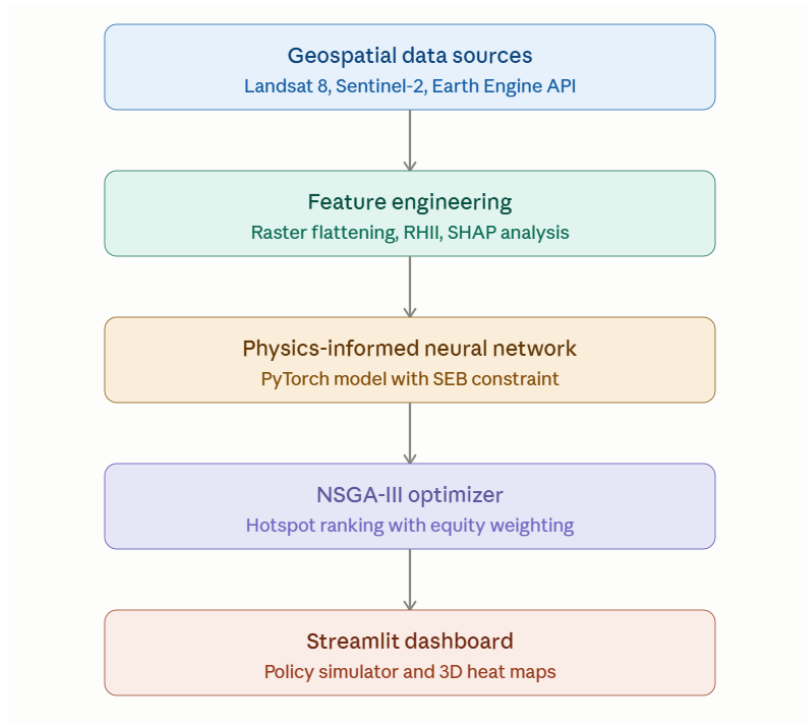
Urban Greening Impact: 1.35 °C

Blue Infrastructure Impact: 0.60 °C

Cool Roofs Impact: 0.60 °C

Suburban Net Temperature Drop: -2.55 °C

Architecture diagram of the proposed solution



Technologies to be used in the solution:

- **Satellite/Geospatial:** Google Earth Engine API, Landsat 8 Collection 2, Sentinel-2 Harmonized, rasterio/GDAL/xarray
- **ML & Explainability:** LightGBM, SHAP
- **Deep Learning/Physics:** PyTorch (custom PINN loss)
- **Optimization:** pymoo (NSGA-II/NSGA-III)
- **Data Science:** NumPy, Pandas, SciPy, scikit-learn
- **Frontend/Visualization:** Streamlit, PyDeck, Plotly, Folium
- **Generative AI:** LangChain + OpenAI API (GPT-4o-mini)
- **Language:** Python 3.10+



Estimated implementation cost (optional):

- **Satellite data licensing:** ₹0 (Google Earth Engine public/research tier; Landsat + Sentinel are public domain)
- **Cloud compute (PINN training):** ~₹12,000-25,000/year (AWS/GCP spot GPU instance for retraining)
- **Generative AI inference (LangChain agent):** ~₹2,000-5,000/month (scales with usage)
- **Dashboard hosting:** ₹0-8,000/year (Streamlit Community Cloud, or basic cloud VM)
- **Total estimated annual cost:** ~₹40,000-80,000/year for full production deployment

Municipal ROI: the 5.42°C dense-core cooling reduces peak summer grid load by an estimated 5-8%, avoiding far larger physical infrastructure costs

“The software stack to run this pipeline costs **less than ₹10,000/month** - a fraction of what a single street-level green corridor project costs municipally.”

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THANK YOU